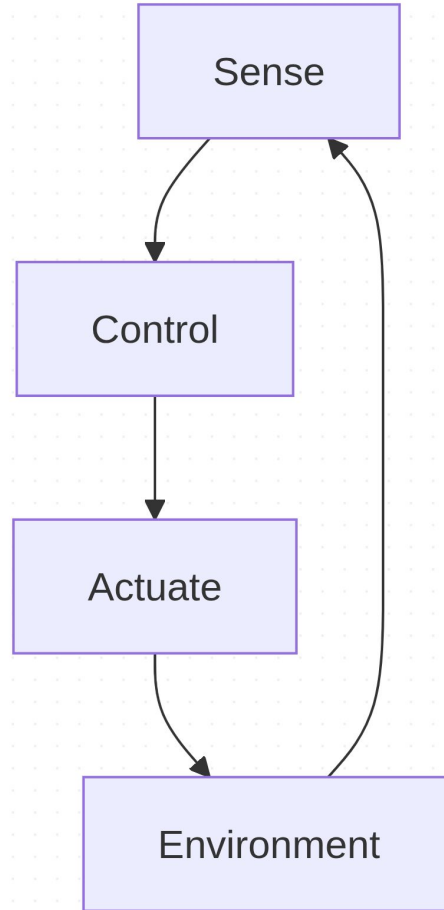


Robotics Hardware

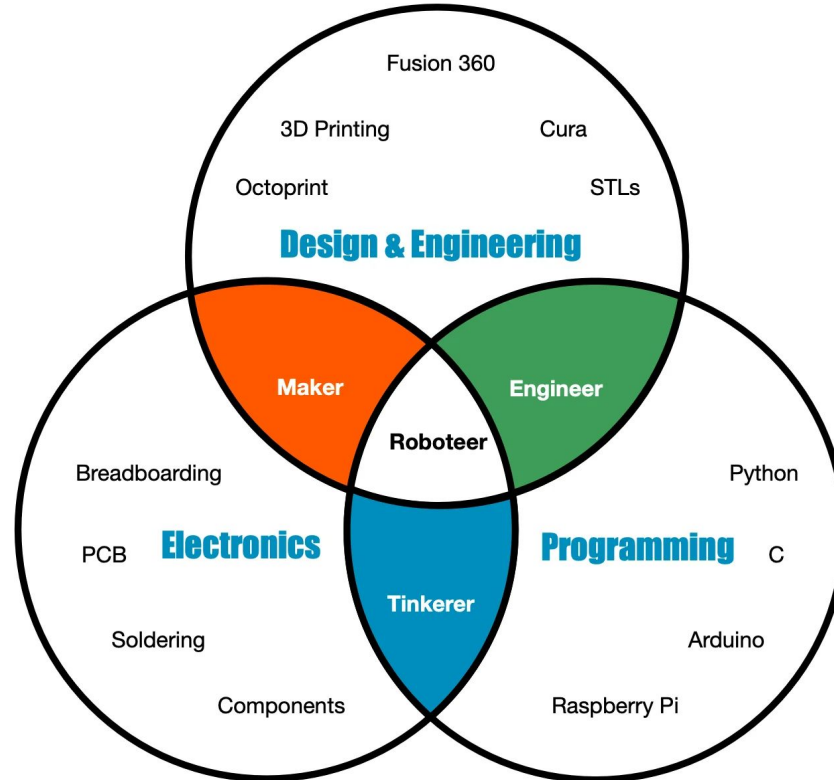
Sensors, Actuators and Systems

What is a Robot?

What is a Robot?



Anatomy of a Robot



Anatomy of a Robot

<https://youtu.be/F29vrvUwqS4>



16:05

I made a DUMMY ROBOTIC ARM from scratch!

1.7M • 4y ago

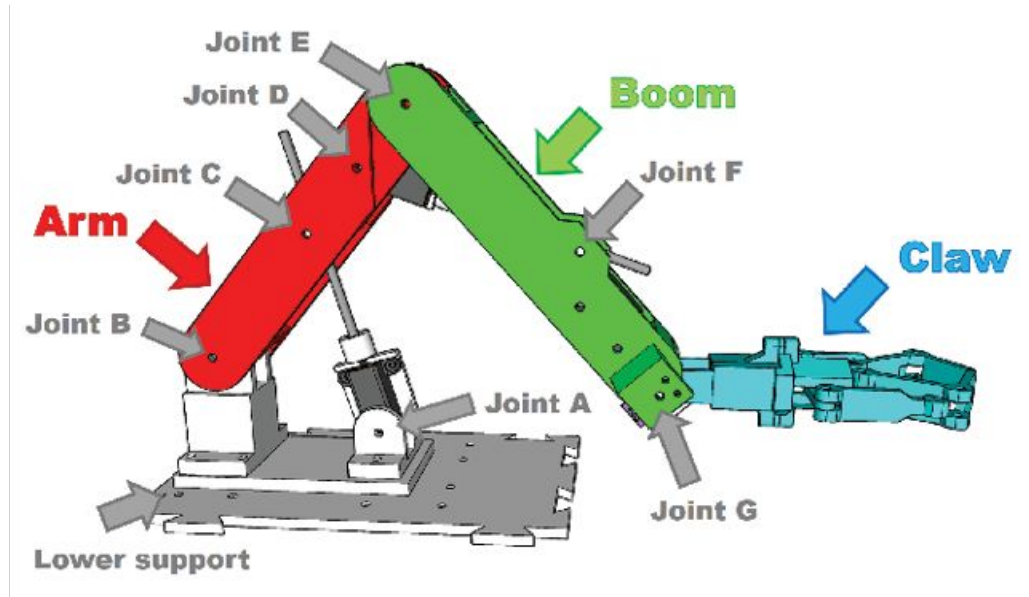


稚晖君

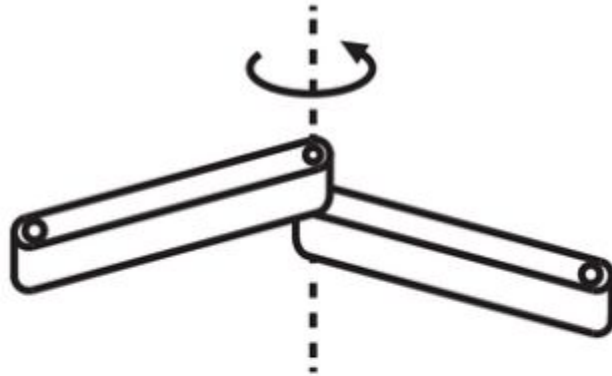
This is the most HARDCORE birthday gift ever that made by myself~ Open-Source repository: ...

CC

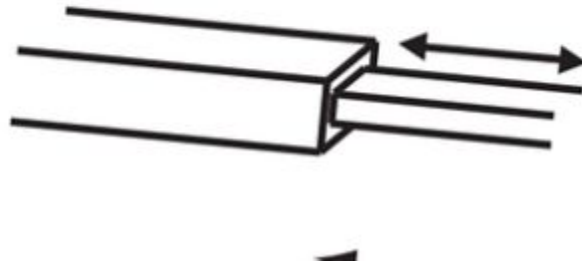
Actuators



What motions do we need to represent?



Revolute
(R)



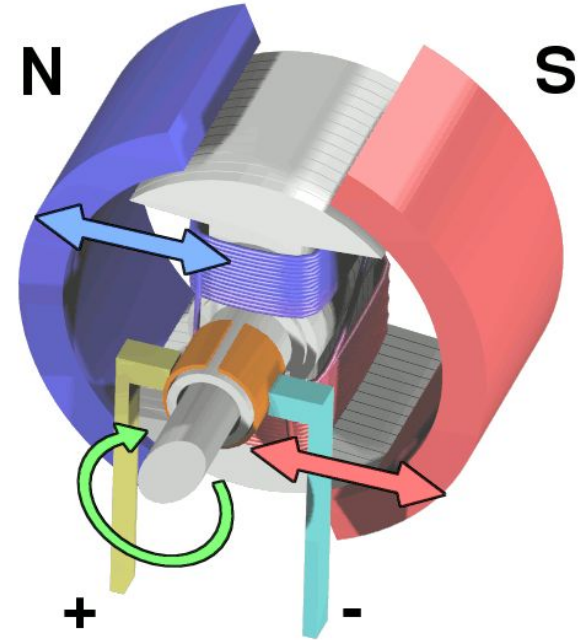
Prismatic
(P)

What is the easiest motion to create?

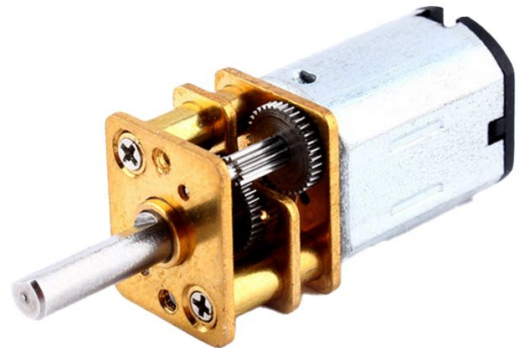
What is the easiest motion to create?

Rotation!

- Torque
- Speed
- Precision



DC Motors



Servo Motors



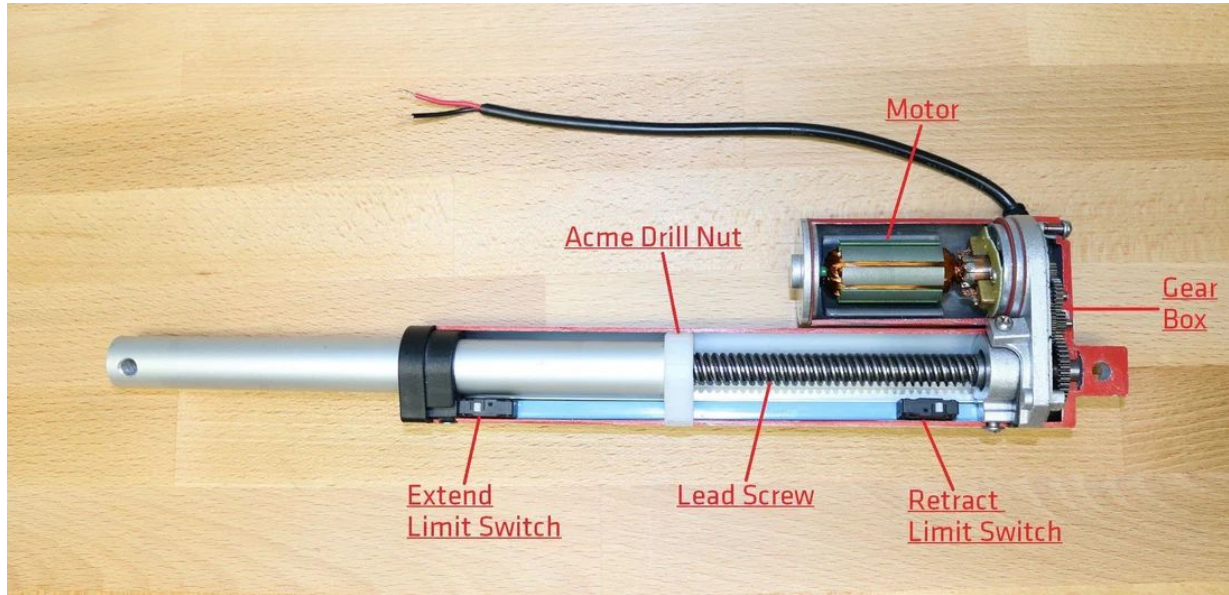
BLDC



What if you want to do something other than spin?

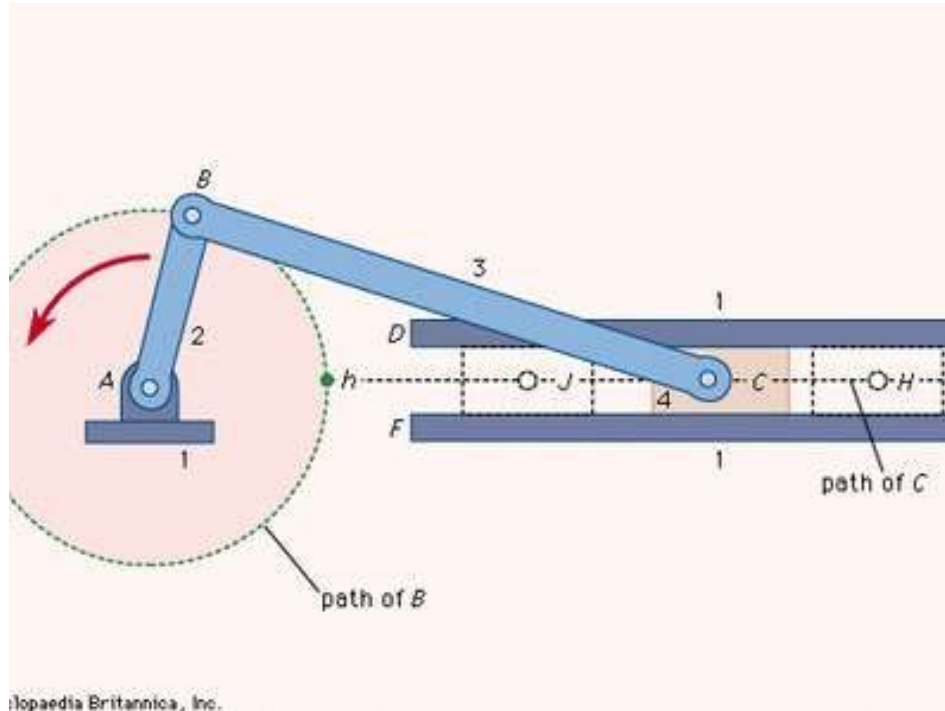
What if you want to do something other than spin?

1. Maybe with a screw?

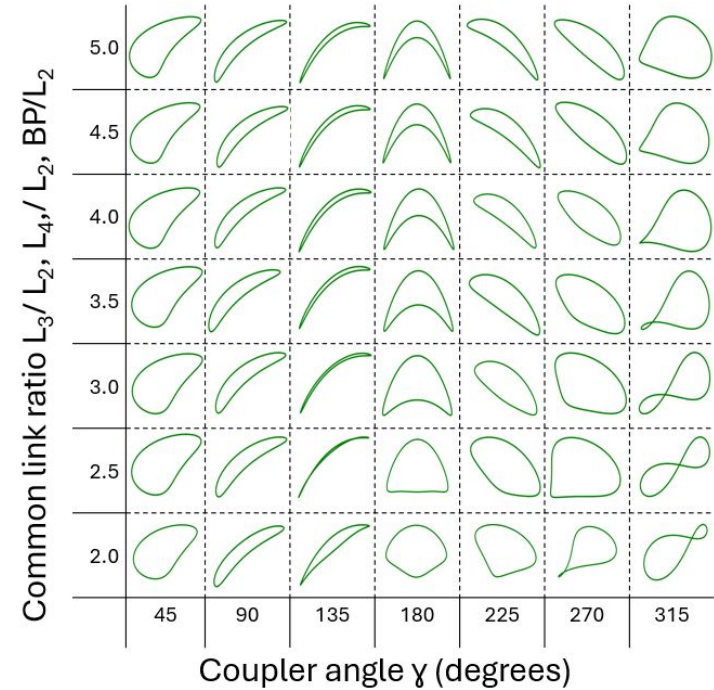
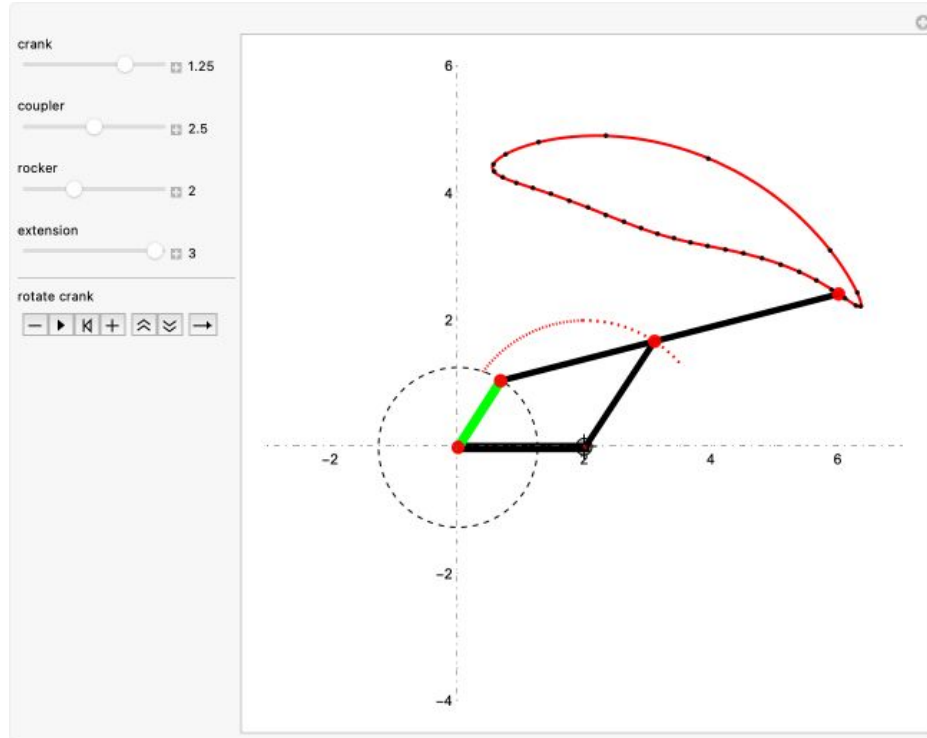


What if you want to do something other than spin?

2. Slider crank mechanism



What if you want even more complex motion?





thang010146

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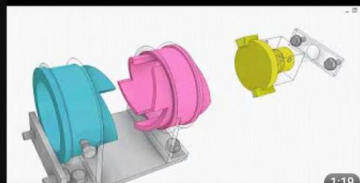
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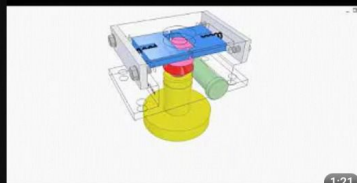


For You



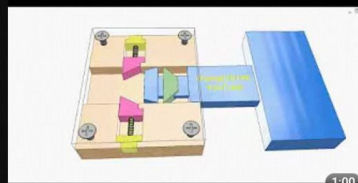
Push-push locking mechanism 2

▷ 12K 6mo ago



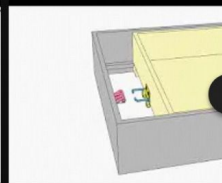
... Device for hanging an object

▷ 3.8K 3w ago



... Push-push locking mechanism 1

▷ 21K 7mo ago



... Push-push locking mechanism 3

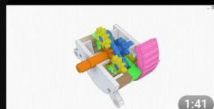
▷ 9.1K 3w ago

Videos



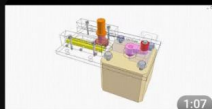
Unstable positions of parallelogram mechanism

▷ 3.1K 2d ago



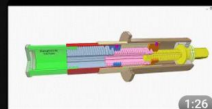
... Two stage gear multiplier

▷ 3.1K 7d ago



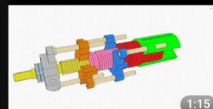
... Planar translation and rotation

▷ 2K 7d ago



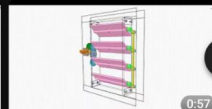
... Screw 3-stage telescopic actuator 2b

▷ 4.3K 12d ago



... Screw 3-stage telescopic actuator 2a

▷ 3.5K 12d ago



... Slat window 4b

▷ 1.5K 12d ago

What actuators does this robot use?



What actuators does this robot use?



Sensors



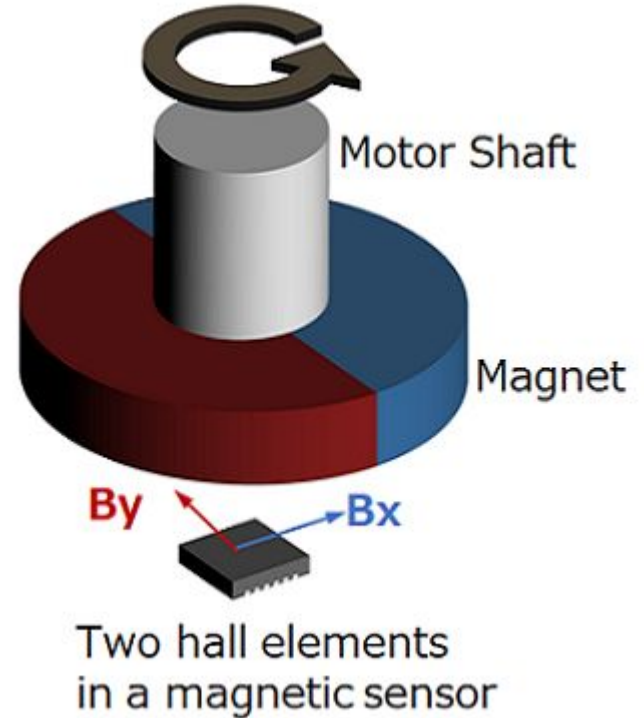
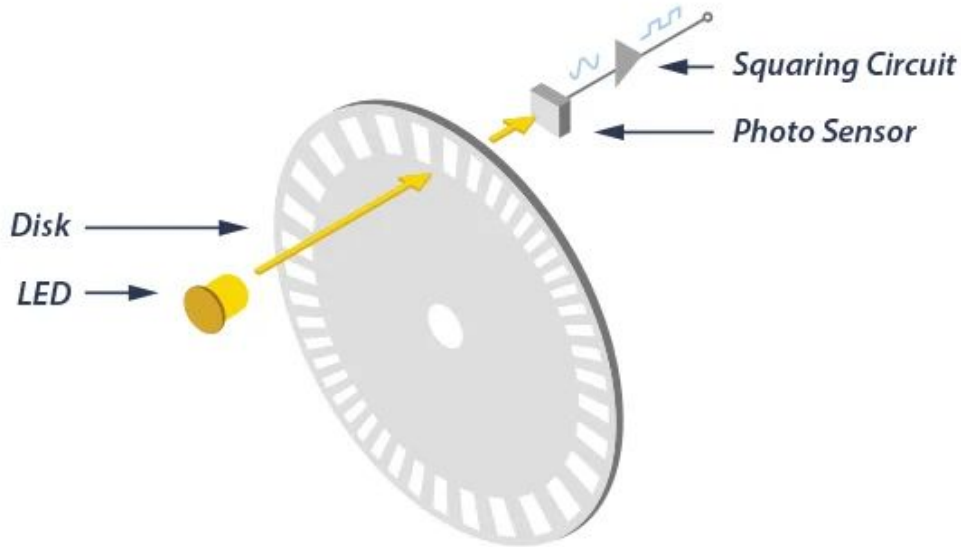
Proprioceptive

Measure internal robot state

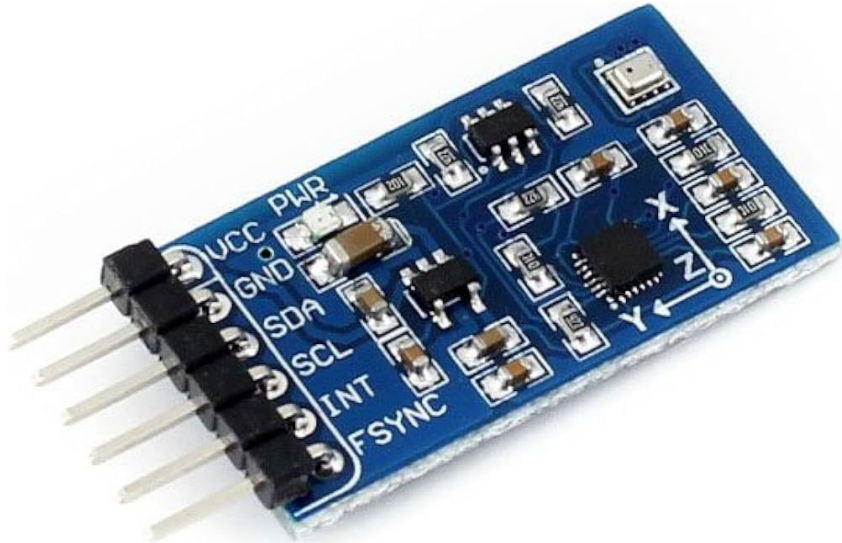
Proprioceptive

Measure internal robot state. But what do we want to measure?

How much has something rotated?



How much/ how fast has something moved?



Other things

- Battery level
- Current draw
- CPU temperature
-

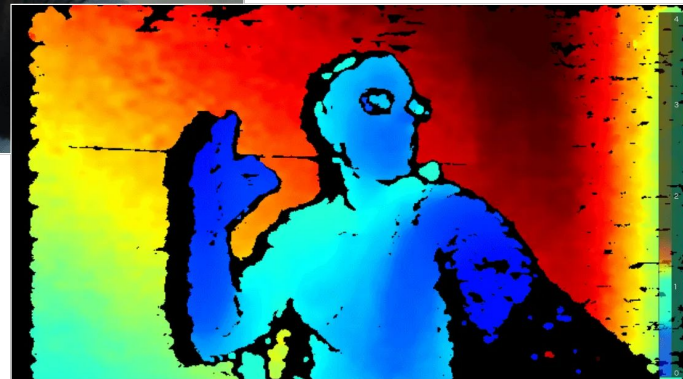
Exteroceptive

Measure environment

Images



DEPTH Images 🤩🤩🤩🤩



Another way to get depth?



(a)



(b)

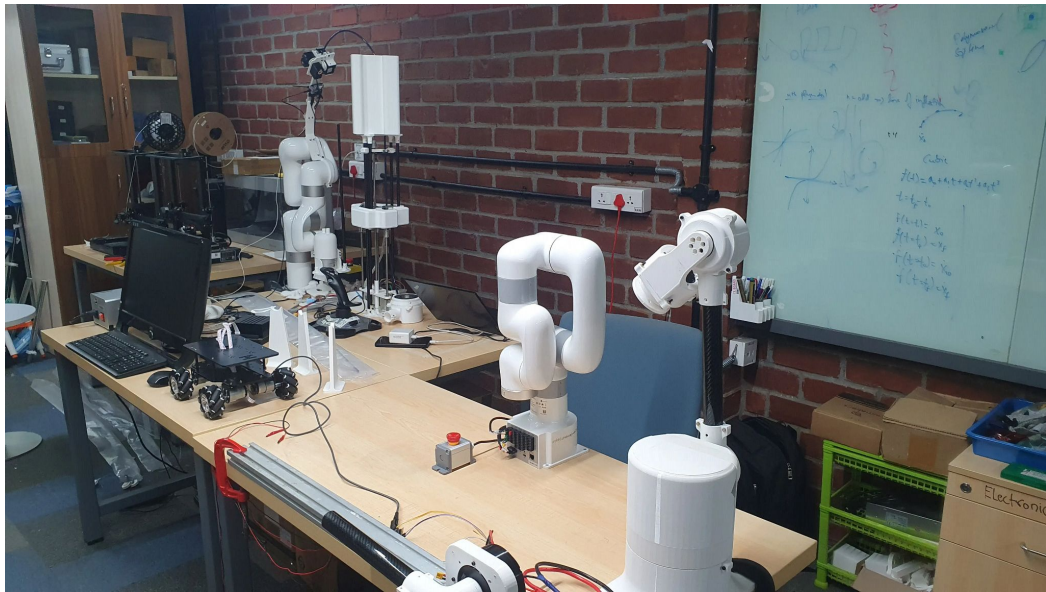
LIDAR

- Longer range
- Outdoor/Indoor



RADAR

Lab Hardware



P3DX



Husky



xArm7 & 6Lite



Lots of Drones!